ITEC 235-940 System Administration

Microsoft DHCP Lab

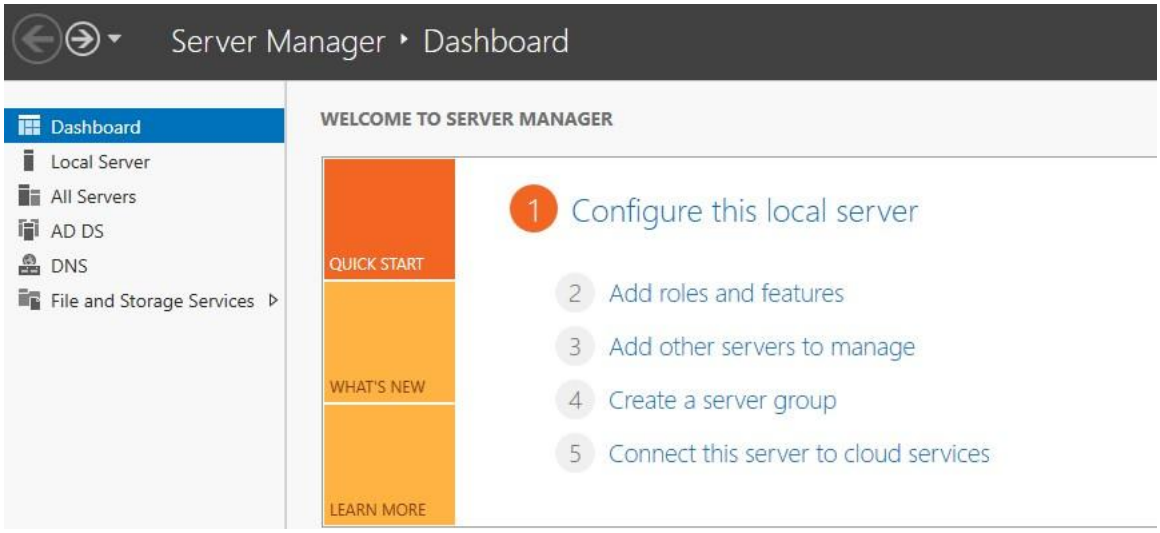
Introduction

The Microsoft Dynamic Host Configuration Protocol (DHCP) server role automatically assigns IP configuration to a network node. DHCP may be installed with Server Manager or PowerShell. This lab provides a brief overview of Microsoft DHCP Server.

Installation of DHCP Server

1. Start and log onto your Microsoft Server 20xx VM.

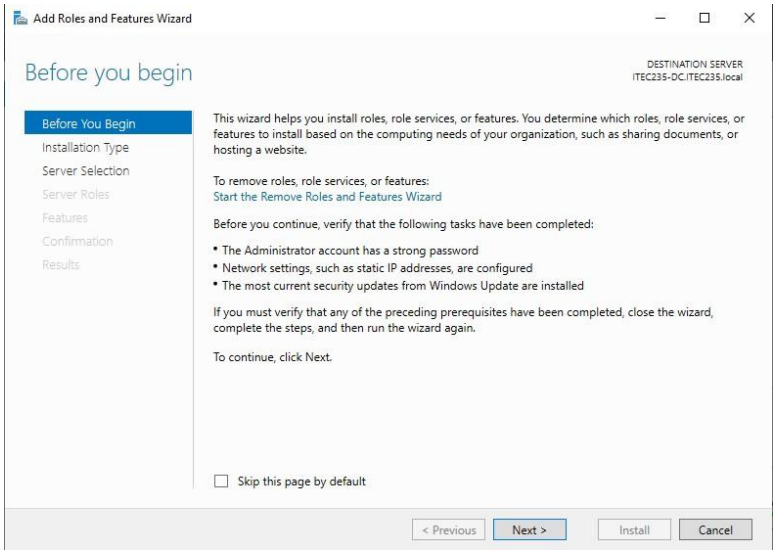
2. Open **Server Manager**.



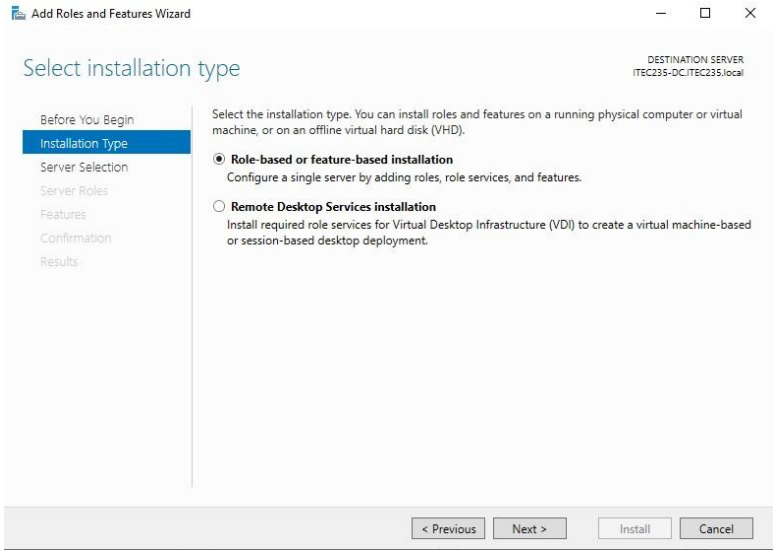
3. Select **Local Server** from the left column.

4. Select **Add roles and features** from the Quick Start menu.

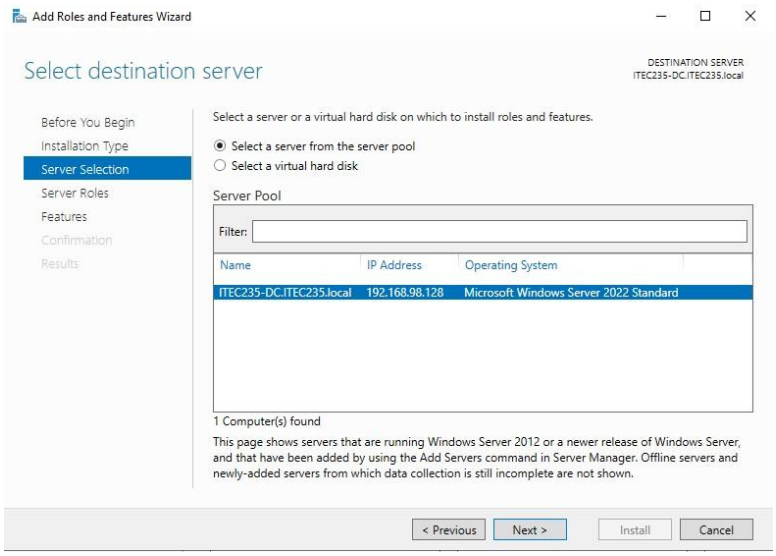
5. Click the **Next** button on the *Before you begin* window.



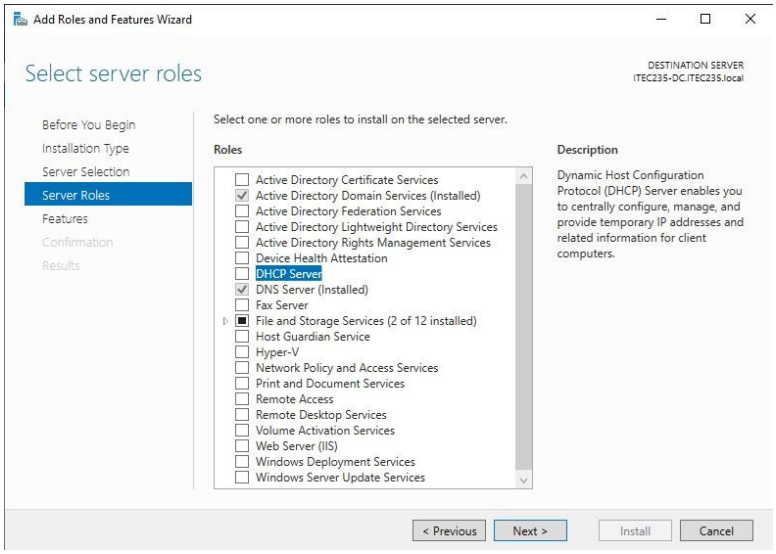
6. Select the **Role-based or feature-based installation** radio button on the *Select installation type* window and click the **Next** button.



7. Select your server from the *Server Pool* and click the **Next** button.

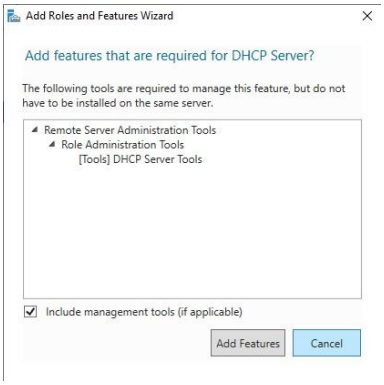


8. Select the **DHCP Server** check box from the list of *Roles*.

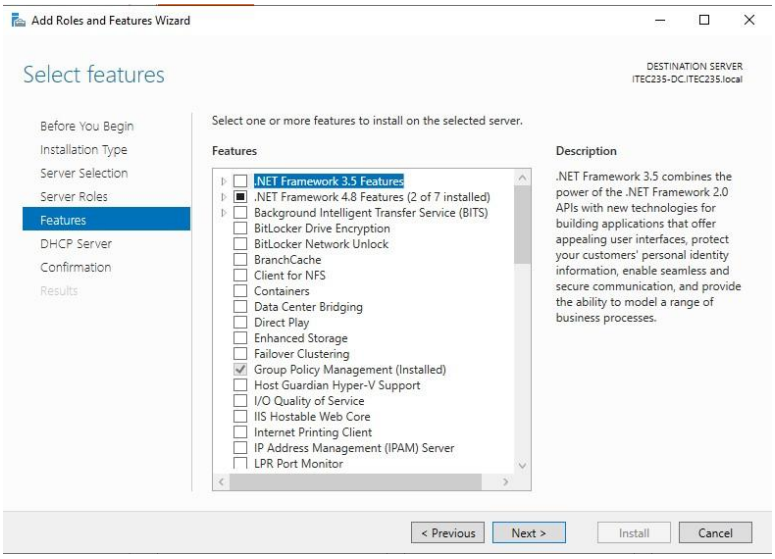


9. Click the **Add Features** button at the bottom of the *Add Roles and Features Wizard* window.

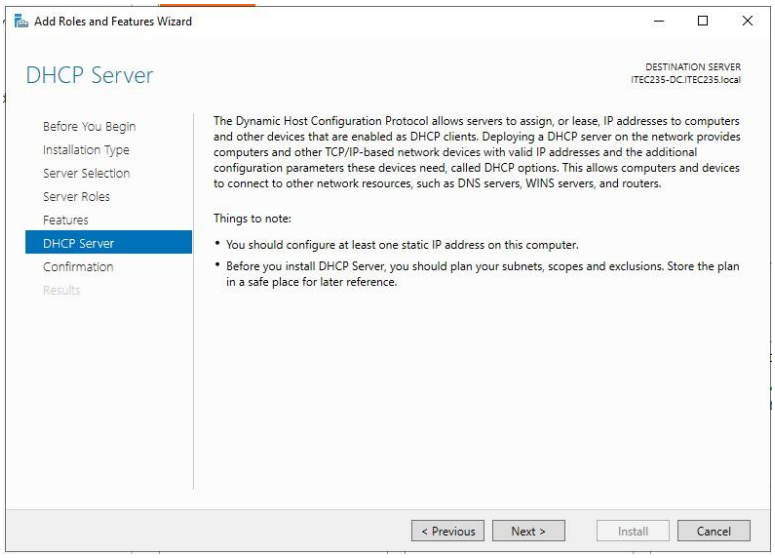
10. Click the **Next** button.



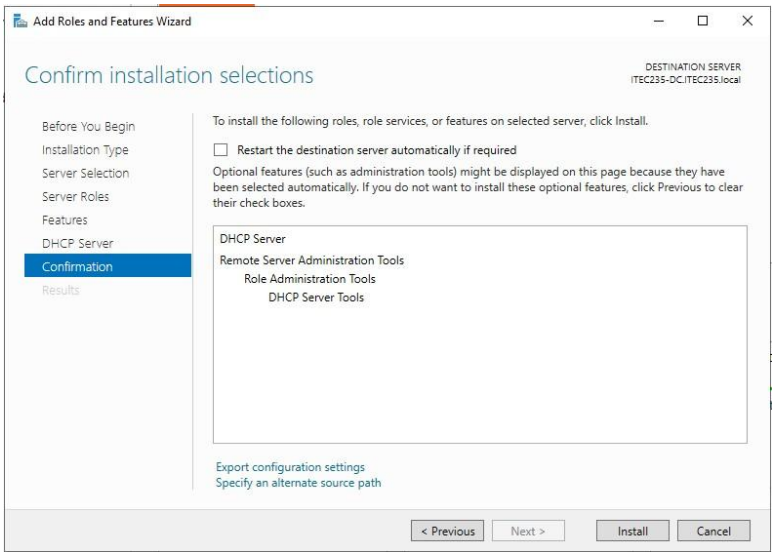
11. No additional features are needed. Click the **Next** button.



12. Click the **Next** button.

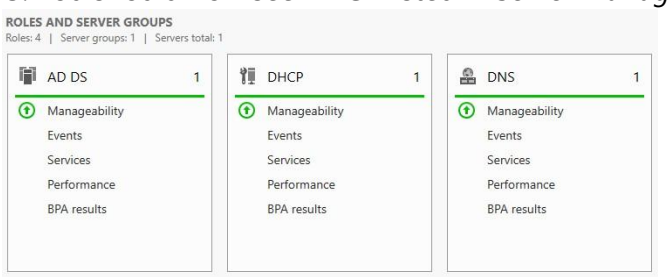


13. Click the **Install** button at the bottom of the *Confirm installation selections* window.

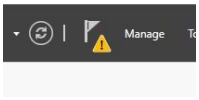


14. Click the **Close** button once the installation of DHCP is complete.

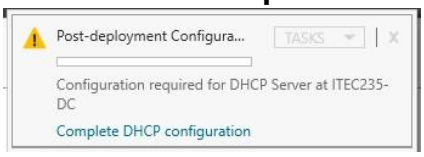
15. You should now see DHCP listed in *Server Manager* under *Roles and Server Groups*.



16. Click on the Server Manager notifications icon.

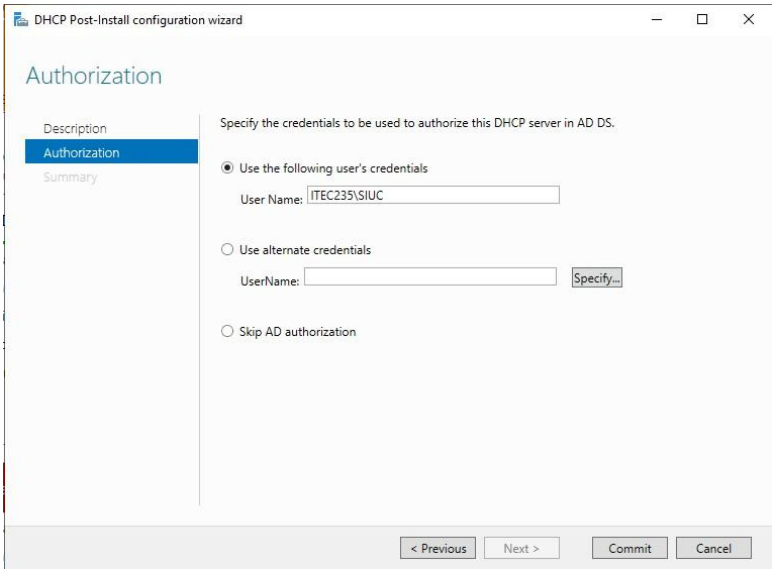


17. Click on the **Complete DHCP configuration** hypertext.



18. Click the **Next** button at the bottom of the *DHCP Post-Install configuration wizard* window.

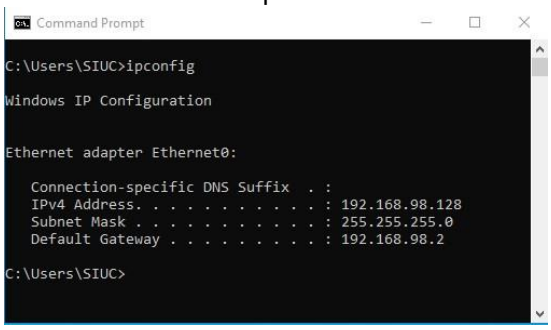
19. Select the *Use the following user's credentials* radio button and enter your username. Click the **Commit** button.



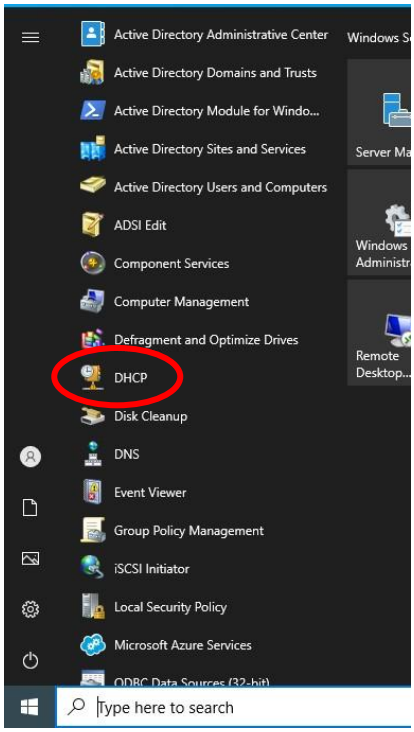
20. Make sure both tasks have completed successfully and click **Done**.

DHCP Server Lab

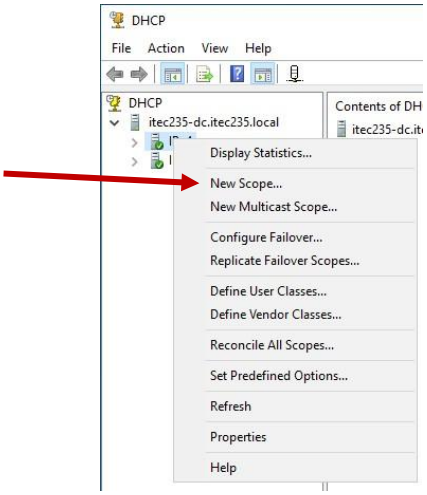
1. Note the IP address, subnet mask, and default gateway for your server. I opened a Command Prompt window and entered the command **ipconfig**.

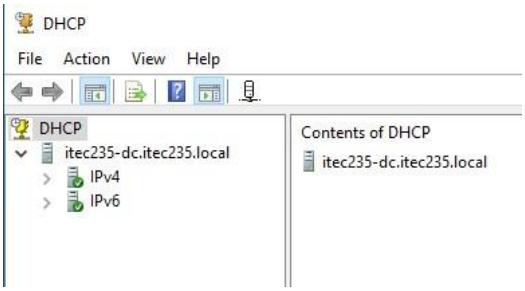


2. Click **Start**, **Windows Administrative Tools**, and then **DHCP**.



3. Expand your server in the column on the left and right-click **IPv4**. Select **New Scope** from the menu.

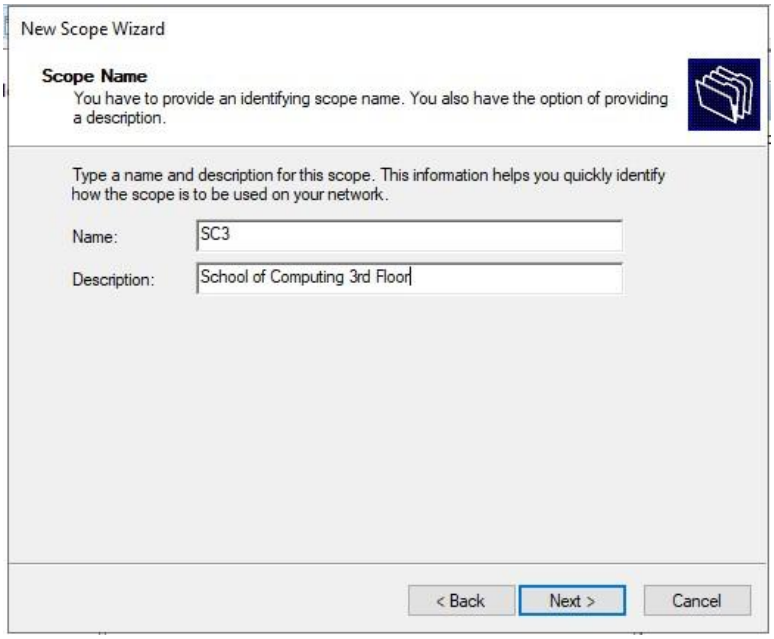




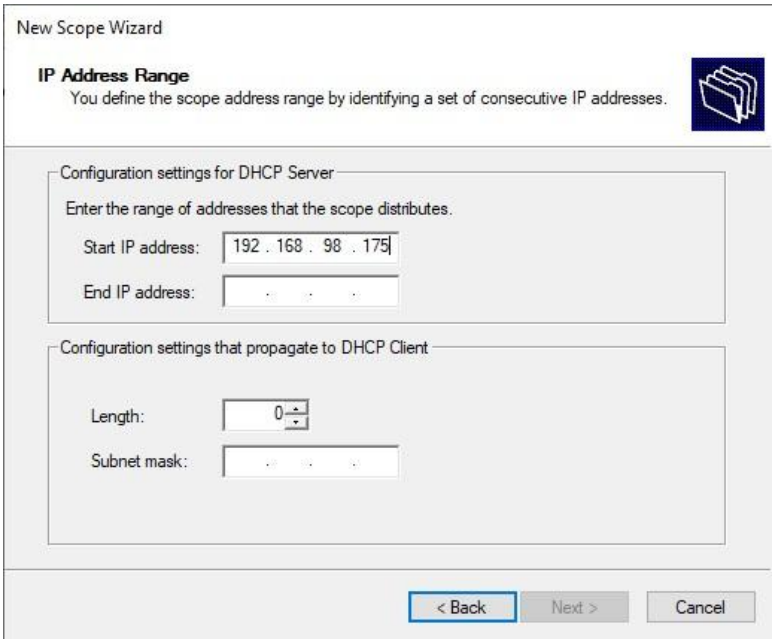
4. Click the **Next** button on the *New Scope Wizard* window.



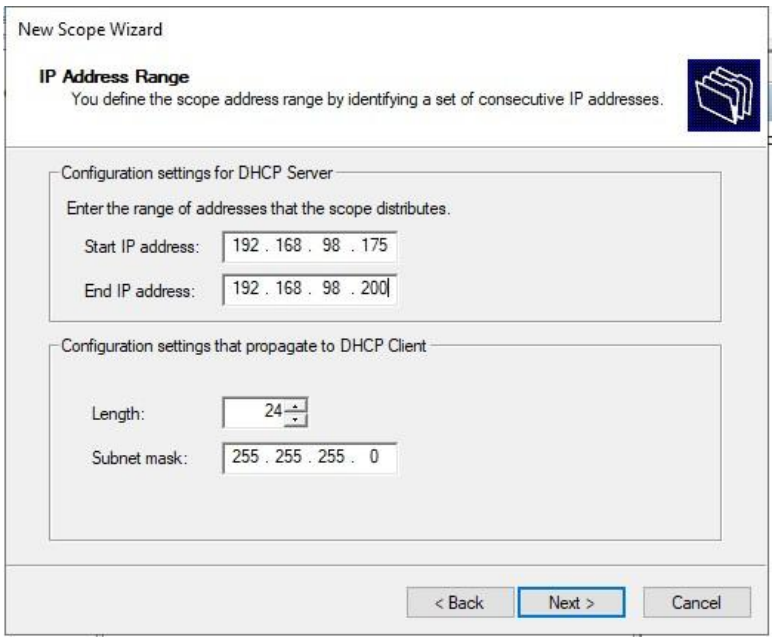
5. Enter a Name and Description and click the **Next** button.



6. In the *Start IP address* field, enter the first three octets of your server's IP address followed by the number **175**. For example, my server's IP address is 192.168.98.128, so I will enter 192.168.98.175 in the *Start IP address* field.

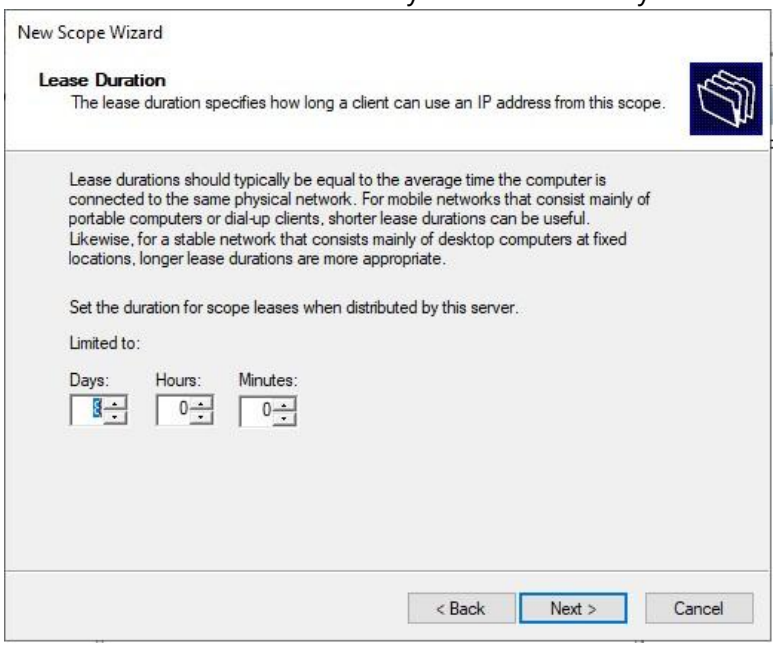


7. In the *End IP address* field, enter the same first three octets as you did in step 6 followed by **200** in the fourth octet. For example, I used the first three octets of 192.168.98 in step 6, so I will enter 192.168.98.200 in the *End IP address* field. Notice that the Subnet mask field is populated automatically. Remember that Microsoft DHCP Server does not support variable length subnet mask (VLSM). Click the **Next** button.

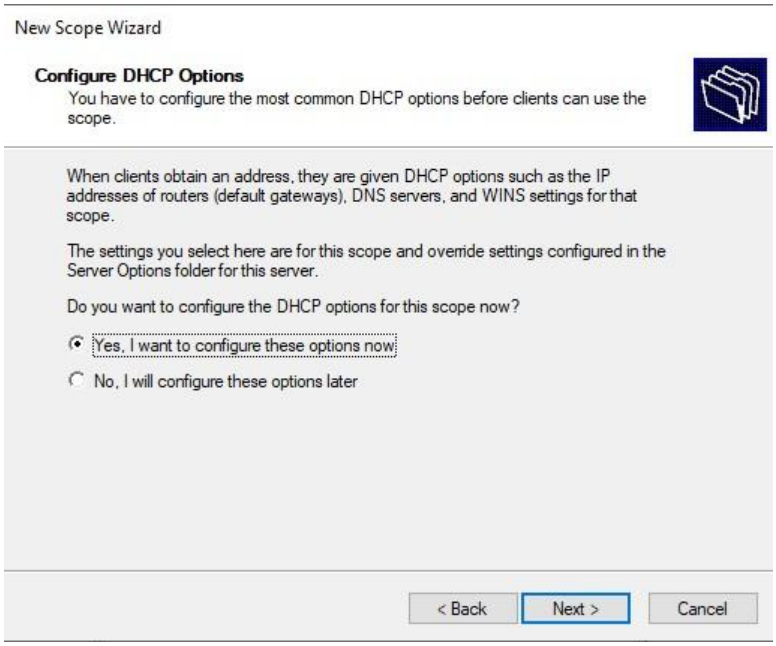


8. We will not add exclusions right now, so click the **Next** button.

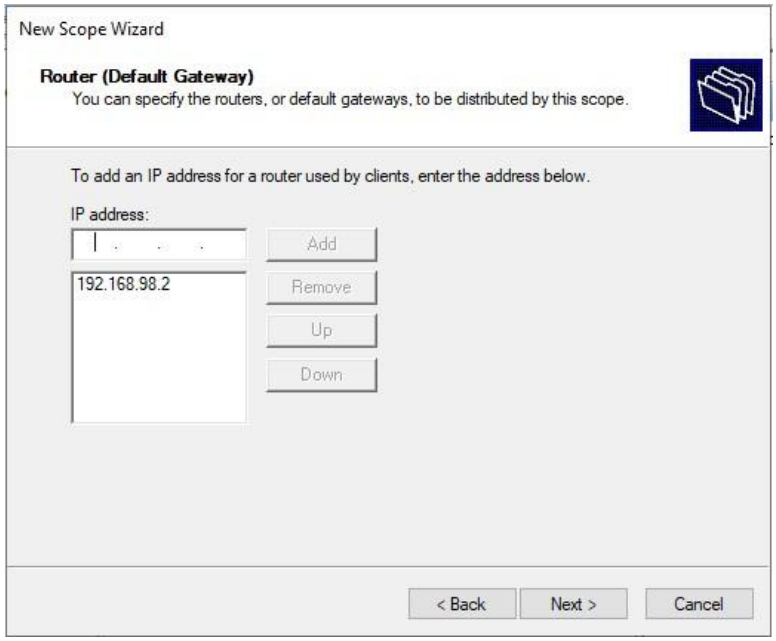
9. Set the lease duration to any amount of time you desire and click the **Next** button.



10. Select *No, I will configure these options later* and click the **Next** button.



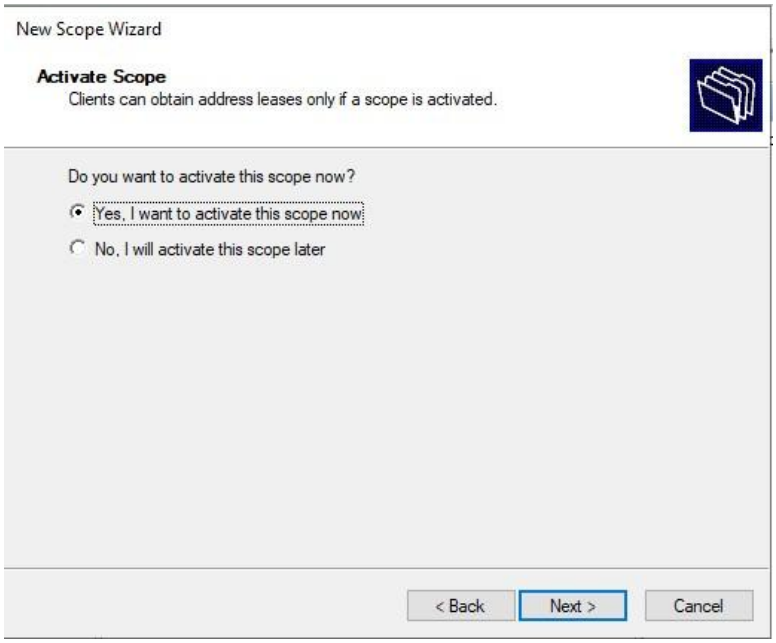
11. Enter the default gateway address as noted during step 1. Click the **Add** button and then the **Next** button.



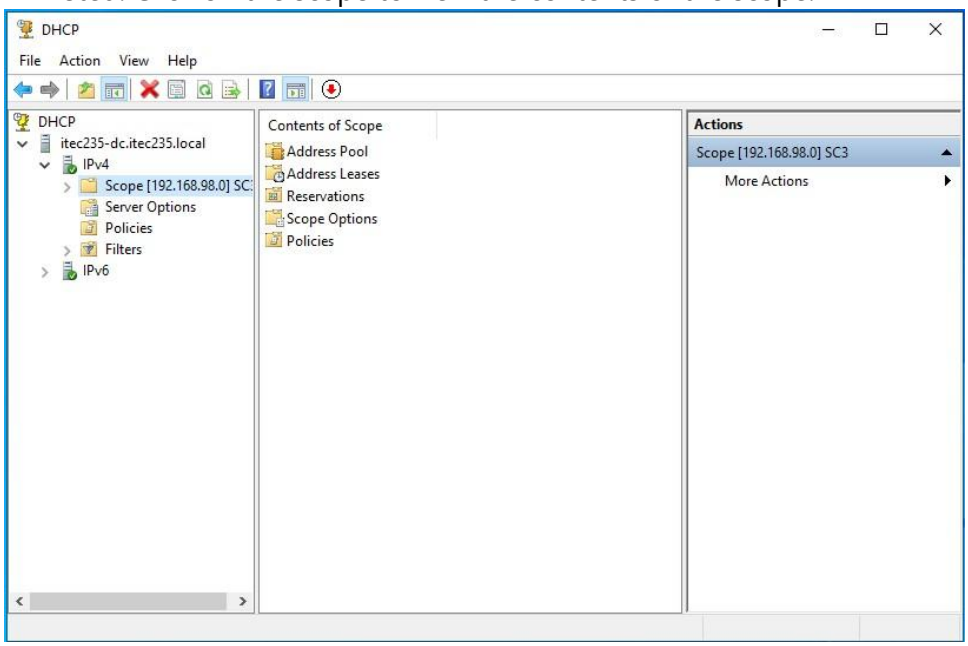
12. Click the **Next** button at the bottom of the Domain Name and DNS Servers window.

13. WINS is an older technology, similar to DNS, that is rarely needed. Click the **Next** button.

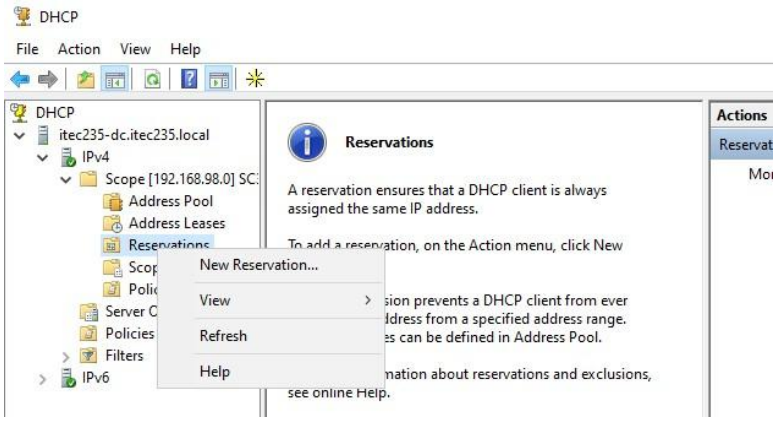
14. Select *Yes, I want to activate this scope now* and then click the **Next** button. Click the **Finish** button to complete the New Scope Wizard.



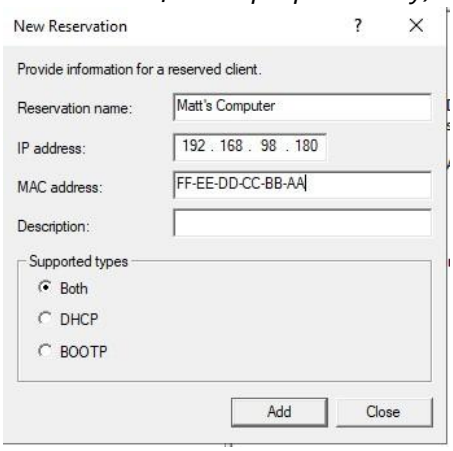
15. Expand IPv4 in the left column of the DHCP snap-in. Notice that you now have a scope listed. Click on the scope to view the contents of the scope.



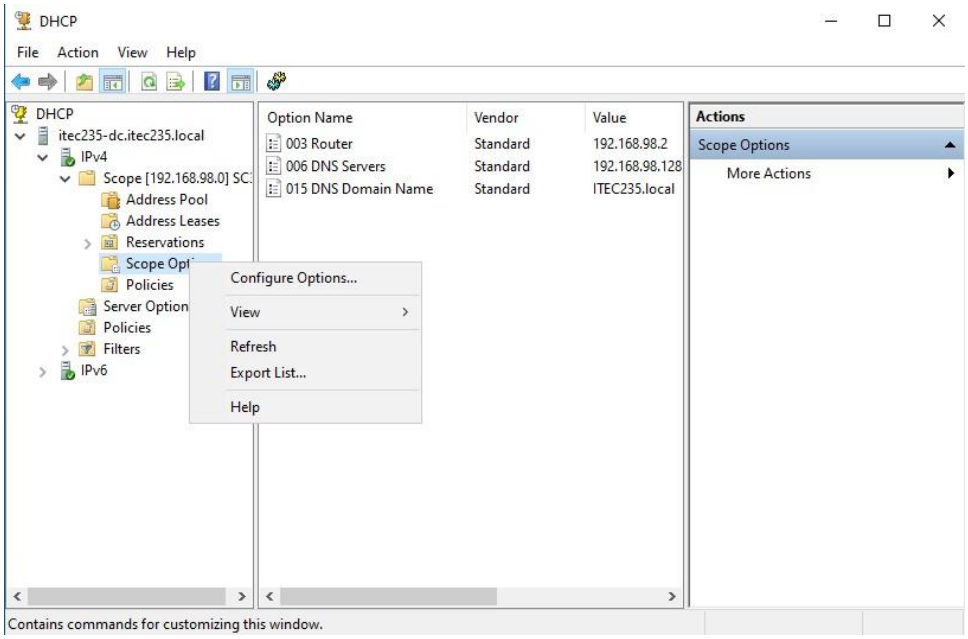
16. Expand the scope. Right click on **Reservations** and select **New Reservation**.



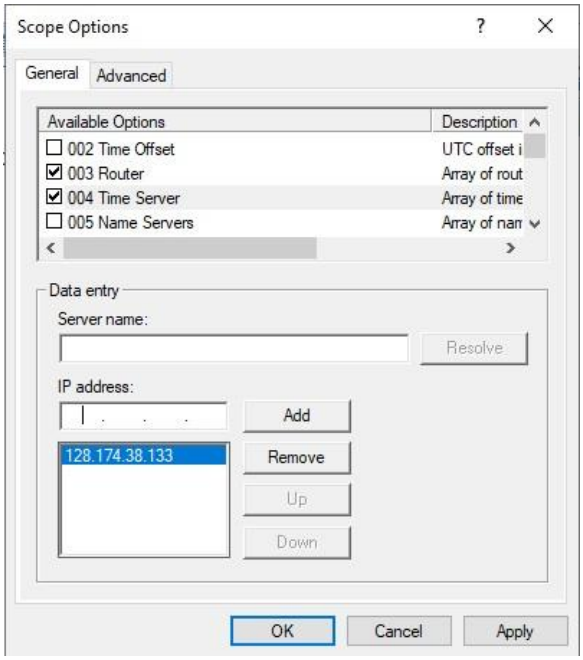
17. Enter a name for your new reservation in the *Reservation name* field. Enter **180** in the fourth octet of the *IP address field*. Enter **FF-EE-DD-CC-BB-AA** (*This is a fake MAC address for lab purposes only)* in the *Mac address* field. Click the **Add** button.



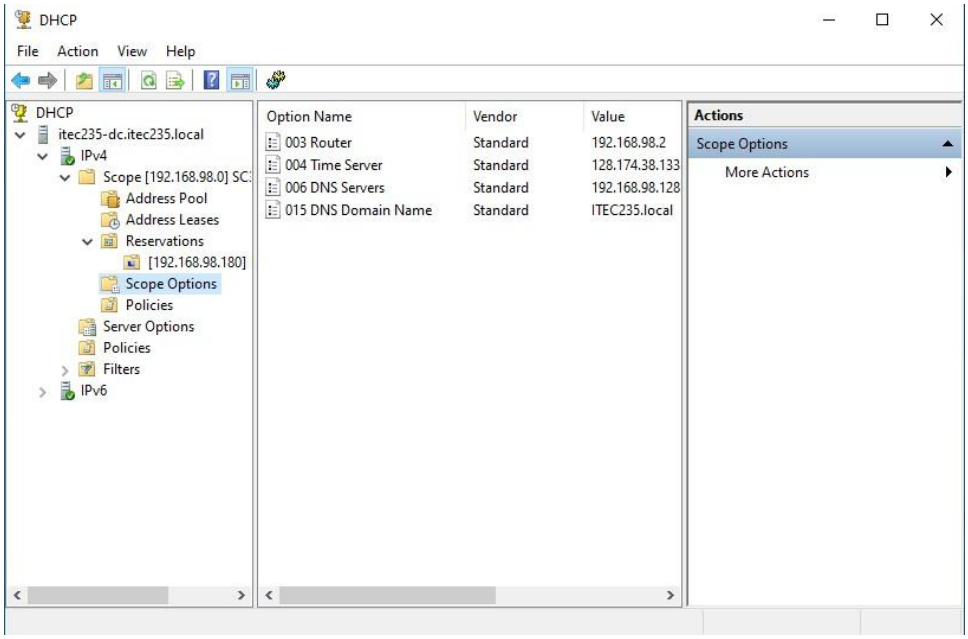
18. Right click *Scope Options* and select **Configure Options**.



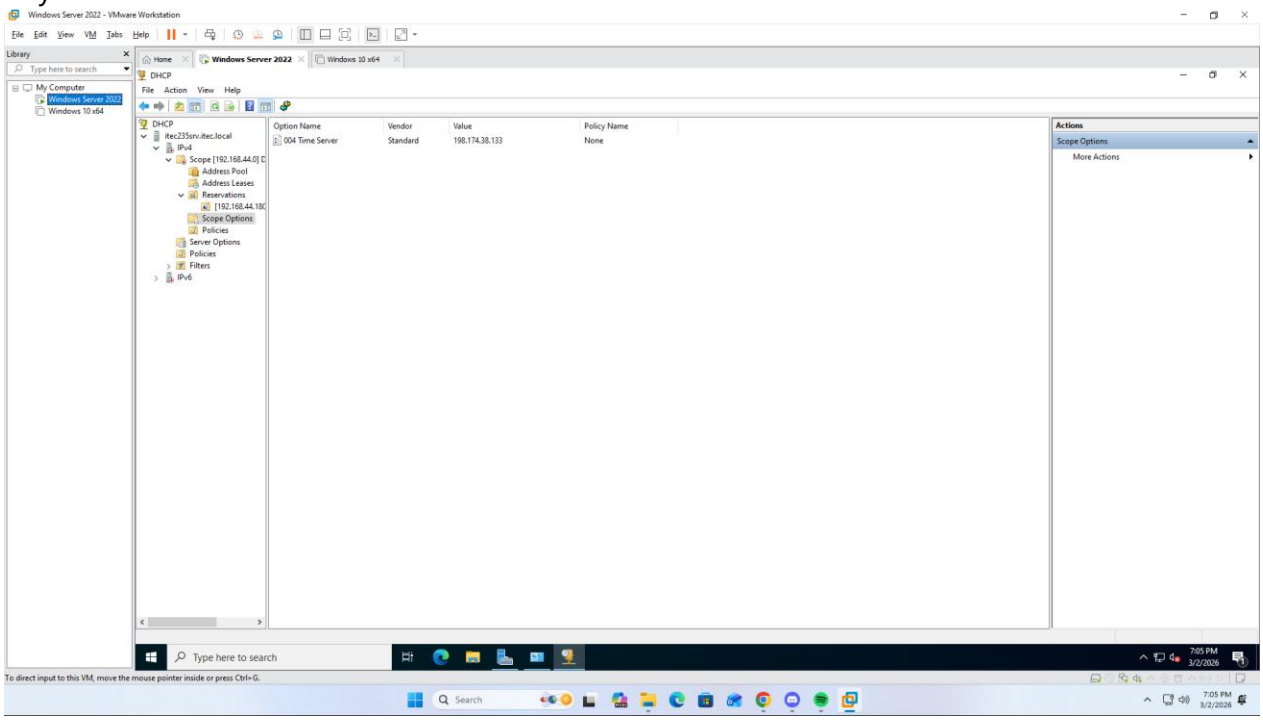
19. Select **004 Time Server**. Enter **Uofl** in the Server name field. Enter **128.174.38.133** in the IP address field and click the **Add** button followed by the **OK** button. This will set the client's time server to UofI.



20. Expand Reservations and click on Scope Options. Then create a screen capture of the DHCP window. Upload your screen capture to the lab submission link. See example below.


DHCP
File Action View Help
DHCP
itec235-dc.itec235.local
IPv4
Scope [192.168.98.0] SC
Address Pool
Address Leases
Reservations
[192.168.98.180]
Scope Options
Policies
Server Options
Policies
Filters
IPv6
Option Name
Vendor
Value
003 Router
Standard
192.168.98.2
004 Time Server
Standard
128.174.38.133
006 DNS Servers
Standard
192.168.98.128
015 DNS Domain Name
Standard
ITEC235.local
Actions
Scope Options
More Actions

My Answer:


Windows Server 2022 - VMware Workstation
DHCP
Option Name
Vendor
Value
Policy Name
004 Time Server
Standard
198.174.38.133
None
Actions
Scope Options
More Actions
Type here to search